1. Project Title

Sree Maruti Agro Kendra – Agri-E-Commerce Platform

---

2. Problem Statement

Sree Maruti Agro Kendra currently operates as a physical retail shop, which limits its customer reach and makes inventory and sales management a manual process. This project aims to create a dedicated e-commerce platform to bring the business online. The platform will solve these issues by enabling online sales, automating inventory management, and expanding the customer base beyond the immediate locality. A key feature will be offering **location-specific product availability and pricing**, catering to the logistical and regional constraints of selling agricultural products.

---

3. System Architecture

The application will follow a classic client-server architecture. The frontend, built with Next.js, will communicate with a backend API, which in turn interacts with the MySQL database.

Frontend (Client) → Backend (REST API) → Database (MySQL)

- **Frontend:** Next.js with JavaScript and CSS for a dynamic, server-rendered user experience.
- **Backend:** Node.js with the Express.js framework to build a robust REST API.
- **Database:** MySQL (Relational) for structured data storage of users, products, and orders.
- **Authentication:** JSON Web Token (JWT) will be used to secure the API and manage user sessions.
- **Hosting:**
    - **Frontend** → Vercel (Optimized for Next.js)
    - **Backend** → Render / Railway
    - **Database** → Railway / PlanetScale / Aiven

---

4. Key Features

Category	Features

Authentication & Authorization	User registration, login, and logout. Role-based access control to distinguish between **Customers** and **Admin** (shop owner).
CRUD Operations	Admin will have full Create, Read, Update, and Delete capabilities for products, categories, and inventory. Users can manage their profiles and view orders.
Frontend Routing	Multi-page navigation including: Home, Products, Product Details, Cart, Checkout, User Profile (with order history), Admin Dashboard, Login, and Signup pages.
Location-Based Services	The system will prompt users for their location (e.g., pincode or district). Based on this, it will **display available products** and show **region-specific pricing**.
Product Discovery	Users can easily find products using advanced **filtering** (by category, brand), **searching** (by name), and **sorting** (by price, name).
Admin Dashboard	A secure area for the shop owner to manage inventory, view sales data, process orders, and update product information with **pagination** for easy management.
Hosting	Both the frontend and backend will be deployed to live, publicly accessible URLs for a complete production-ready application.

---

5. Tech Stack

Layer	Technologies
Frontend	Next.js, React.js, CSS (in separate `.css` files), Fetch API / Axios
Backend	Node.js, Express.js
Database	MySQL, Sequelize (ORM for easier database interaction)
Authentication	JSON Web Tokens (JWT), bcrypt.js (for password hashing)
Hosting	Vercel (Frontend), Render / Railway (Backend & Database)

---

6. API Overview

Here is a sample list of REST API endpoints that will be implemented.

Endpoint	Method	Description	Access
`/api/auth/register`	`POST`	Registers a new customer.	Public
`/api/auth/login`	`POST`	Authenticates a user and returns a JWT.	Public
`/api/products`	`GET`	Fetches all products. Supports query params for location, search, sort, filter, and page.	Public
`/api/products/:id`	`GET`	Fetches details for a single product.	Public
`/api/products`	`POST`	Creates a new product.	**Admin only**
`/api/products/:id`	`PUT`	Updates an existing product's details.	**Admin only**
`/api/products/:id`	`DELETE`	Deletes a product from the database.	**Admin only**
`/api/orders`	`POST`	Allows an authenticated user to place a new order.	Authenticated User
`/api/orders/user`	`GET`	Fetches the order history for the logged-in user.	Authenticated User
`/api/orders/all`	`GET`	Fetches all customer orders for the admin.	**Admin only**

Name: Shrihari K N
Enroll no: 2401010448
Batch: Charles Babbage 1